OpenVLA Thesis-Style Master Report

A full explanation of the project, written for a professor, thesis committee, and future self

Prepared from the completed local OpenVLA experiments and submission bundle

Includes project motivation, research gap, debugging history, methods, results, limitations, and future work

What this file is for
This package explains the full OpenVLA project from motivation to debugging, experiment design, results, limitations, and next steps. It is written so a beginner can follow it, while still preserving enough technical detail for a thesis defense or conference discussion.

Executive Summary

This project studied whether OpenVLA-7B changes its predicted robot action when the same scene is paired with different prompt variants. The work used two completed experiments: a manually curated 40-trajectory BridgeData subset and an automatically sampled 50-trajectory subset from the larger scripted BridgeData archive. The central finding was consistent: paraphrased prompts stayed closest to the normal prompt, while contradictory and neutral prompts caused larger action drift. Final-state images were generally more fragile than initial-state images. These results support the thesis that language perturbations measurably affect VLA control outputs and that scene stage matters. [1][2]

One-sentence version: OpenVLA did not behave identically across prompt variants; prompt wording changed action predictions, and final images tended to be more sensitive than initial images.

1. Project Background and Why This Topic Matters

Robot learning is moving from hand-engineered pipelines toward models that combine perception, language understanding, and action generation inside one system. OpenVLA is one of the most important open-source examples of this shift because it packages a large vision-language-action model into a form that researchers can run and fine-tune locally. [1] LeRobot, meanwhile, shows how the broader robotics ecosystem is increasingly focusing on open tools, datasets, and reproducible pipelines, which lowers the barrier to entry for students and labs. [3]

The research gap behind this project is simple but important: many VLA papers show impressive success on benchmark tasks, but far fewer studies ask how stable the action output is when the language instruction is changed in meaning-preserving or meaning-changing ways. In practice, robots deployed in real environments will receive prompts from different users, different phrasings, and possibly incomplete or neutral instructions. A stable system should respond consistently to paraphrases, while still showing meaningful differences under contradiction. [4]

This gap matters for both science and deployment. If the action vector changes strongly under superficial wording changes, then the model may be too brittle for real human use. If the model ignores contradictory prompts, then it may be over-relying on the image and under-using language. If final-state scenes produce more drift than initial-state scenes, that suggests the model behaves differently when a task is already partially or largely completed. These are directly relevant questions for safety, interpretability, and human-robot interaction. [1][4]

2. What Was Actually Built

- A local WSL-based OpenVLA-7B inference environment using 4-bit loading on an RTX 2060 laptop GPU.
- A patched inference path to fix the attention-mask mismatch that prevented successful local prediction.
- A manual 40-trajectory BridgeData study with paired initial and final frames, task-specific prompts, and CSV-based evaluation.
- An automated 50-trajectory study built from the downloaded scripted BridgeData archive, with metadata, prompt families, and batch inference.
- Analysis scripts that generated summary tables, bootstrap confidence intervals, family breakdowns, and failure-case reports.

3. Research Questions and Hypotheses

1. Does OpenVLA keep the action close to the normal prompt when the instruction is paraphrased?
2. Does the model move farther away when the instruction is contradictory or neutral?
3. Are final-state images more fragile than initial-state images?
4. Do the same qualitative patterns hold in both a manual task-specific experiment and a broader automatically sampled validation set?

Main hypothesis: paraphrases should preserve behavior best; contradictory and neutral prompts should increase drift; final states should be more fragile than initial states. This hypothesis is consistent with the broader VLA literature, which treats language grounding as a key part of action generation rather than a decorative side channel. [1][4]

4. Related Work

OpenVLA is a 7B open-source VLA model trained on a large set of real robot demonstrations and designed to support fine-tuning for new tasks. The main promise is that instead of training a new robot controller from scratch, researchers can adapt a pretrained model to new environments and skills. [1] BridgeData V2 is a large manipulation dataset with 60,096 trajectories across 24 environments, designed to support scalable robot learning and generalization. [2] LeRobot shows the current robotics trend toward open, end-to-end stacks for data collection, model training, and deployment. [3] The broader VLA survey literature frames VLA systems as the union of vision, language, and action inside a single tokenized or fused framework, while also highlighting open problems such as real-time control, action representation, robustness, and safety. [4]

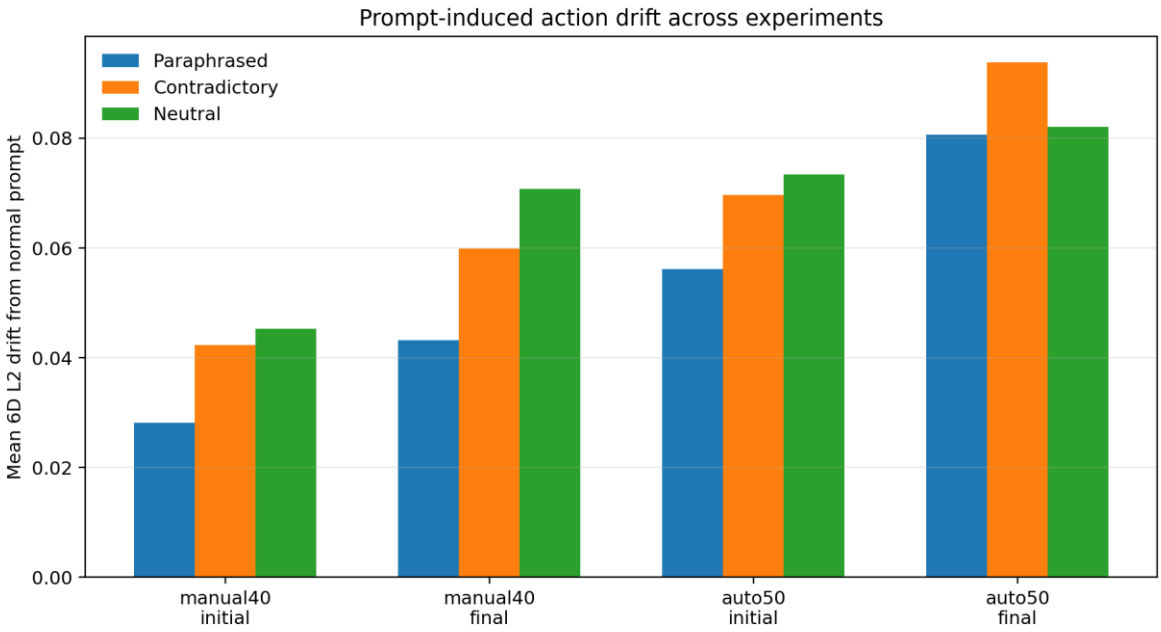

Figure 1. Mean 6D L2 drift across the completed experiments.

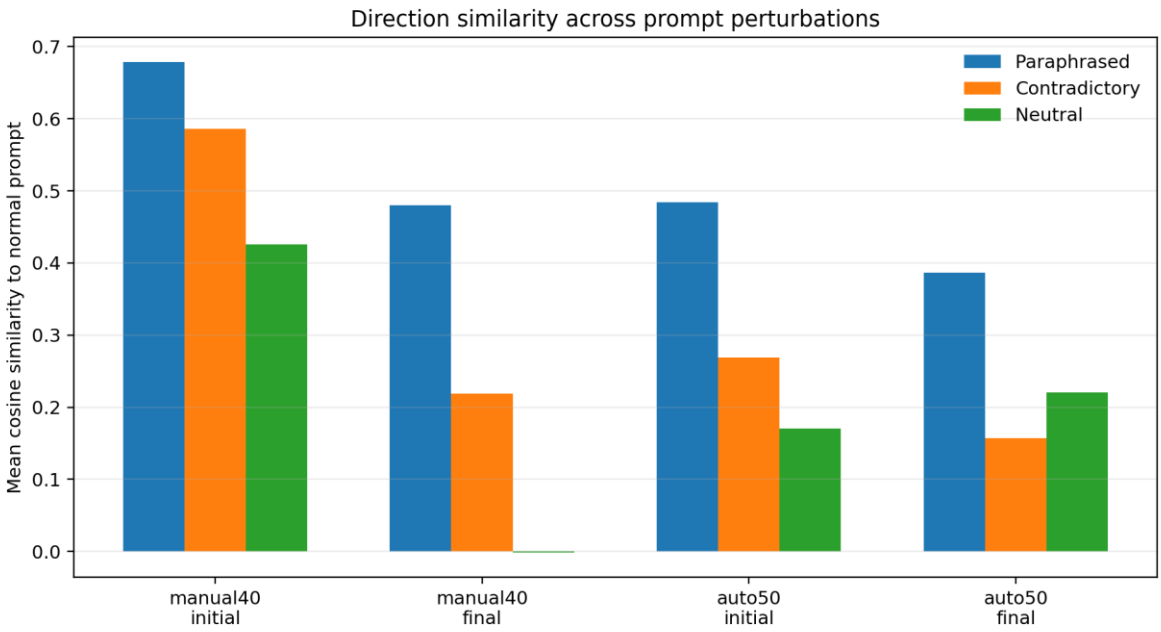

Figure 2. Mean cosine similarity to the normal prompt.

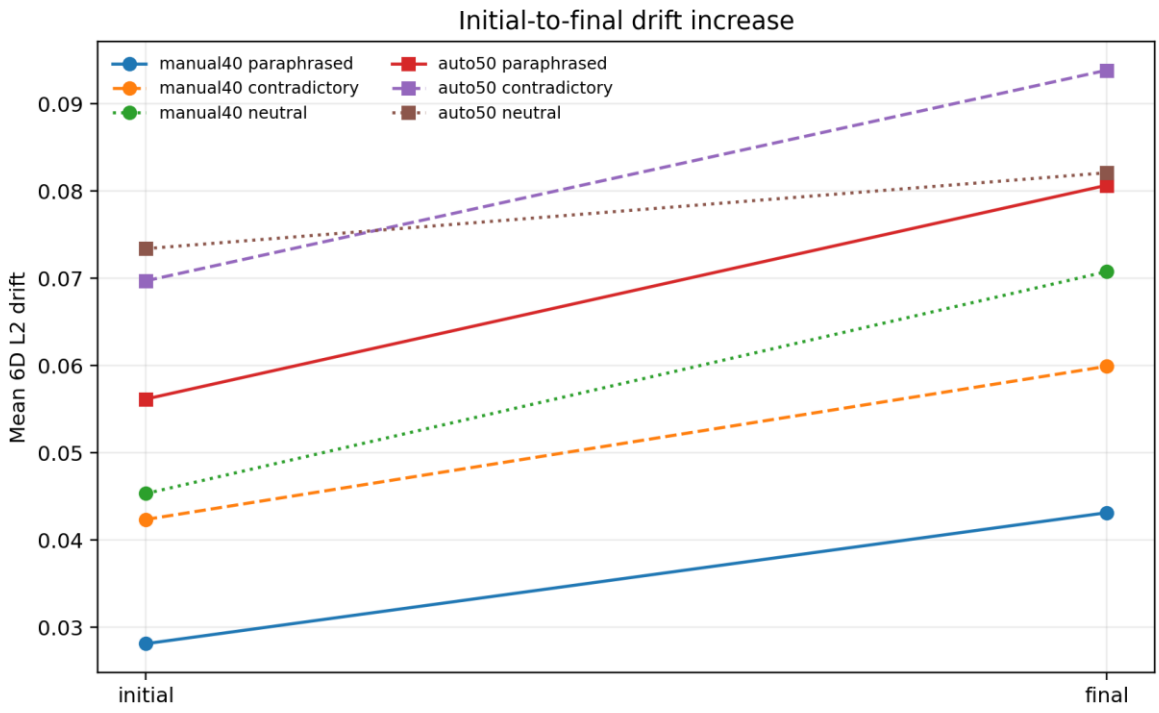

Figure 3. Initial-to-final drift increases by experiment and prompt family.

5. Environment, Setup, and Debugging Story

The final working environment was local rather than cloud-hosted. Early attempts using notebook environments such as Kaggle and similar cloud setups produced dependency and import failures, including missing or incompatible support for the remote-code model class path, bitsandbytes installation issues, and network-related package resolution problems. These failures were useful because they clarified that the project needed a fully controlled environment rather than an ad hoc notebook stack.

The project then moved to local WSL on Ubuntu 22.04.5 with an NVIDIA GeForce RTX 2060. The working environment used Python 3.10 and a dedicated virtual environment. Core packages included PyTorch 2.2.0+cu121, torchvision 0.17.0, torchaudio 2.2.0, transformers 4.40.1, tokenizers 0.19.1, timm 0.9.10, and sentencepiece 0.1.99. The setup successfully detected CUDA and loaded the model in 4-bit mode.

A key turning point was discovering that the model could load but failed during prediction because the text-only attention mask did not match the longer sequence after image tokens were fused into the model input. The practical fix was to patch predict_action so a 256-token image attention mask was prepended before generation. This was not a cosmetic change. Without it, the local OpenVLA test could not run reliably. After the patch, both dummy and real-image inference worked.

This debugging history matters because it shows the project was not a superficial copy of a repository. The work included real systems troubleshooting, environment stabilization, model patching, and reproducible verification. For a thesis defense, this is evidence that the student understands both the scientific question and the engineering path needed to answer it.

Key setup facts

Item	Value
OS / runtime	Ubuntu 22.04.5 under WSL2
GPU	NVIDIA GeForce RTX 2060 (6 GB VRAM)
Model	OpenVLA-7B in 4-bit mode
Main patch	Prepend 256 image attention-mask tokens before predict_action
Prompt families	Normal, paraphrased, contradictory, neutral
Primary metrics	6D L2 drift and cosine similarity relative to the normal prompt

6. Data Preparation and Experiment Design

Experiment 1 (manual40) used 40 trajectories manually curated from BridgeData. For each trajectory, the study saved both the initial frame and the final frame, yielding 80 images. Each image was paired with four prompts: normal, paraphrased, contradictory, and neutral. This branch was task-specific because the language annotations were hand-cleaned and aligned with the particular trajectories.

Experiment 2 (auto50) used the locally downloaded scripted BridgeData archive. A 50-trajectory sample was extracted and expanded into 100 image rows (50 initial + 50 final). Prompts were filled from run-family metadata rather than hand-authored per scene. This branch is broader and more validation-oriented, because it samples multiple family types and tests whether the manual findings survive on a more automated pipeline. [2]

The evaluation intentionally focused on the first six action dimensions. For each image, the normal prompt served as the reference. Drift was measured using L2 distance from the normal action, while directional alignment was measured using cosine similarity to the normal action. Additional counts tracked near-invariant cases and which perturbation was closest or farthest on a per-example basis.

Experiment structure

Branch	Trajectories	Images	Prompt	Main purpose

			variants per image	
manual40	40	80	4	Task-specific primary study
auto50	50	100	4	Broader validation study

7. Main Results

Summary metrics from the completed experiments

Experiment	Group	n	L2 para	L2 contra	L2 neutral	Cos para	Cos contra	Cos neutral
manual40	initial	40	0.028126	0.042365	0.045330	0.678776	0.586046	0.425675
manual40	final	40	0.043141	0.059920	0.070772	0.480281	0.219040	-0.001856
auto50	initial	50	0.056164	0.069716	0.073405	0.483856	0.269209	0.170514
auto50	final	50	0.080668	0.093857	0.082103	0.386891	0.157202	0.220440

The manual40 study produced a clean ordering in both initial and final scenes. In the initial group, mean drift was 0.028126 for paraphrases, 0.042365 for contradictions, and 0.045330 for neutral prompts. In the final group, those values rose to 0.043141, 0.059920, and 0.070772. This supports the claim that paraphrases preserve behavior best and that final scenes are more fragile.

The auto50 validation study preserved most of the story. In the initial group, mean drift was 0.056164, 0.069716, and 0.073405. In the final group, mean drift was 0.080668, 0.093857, and 0.082103. Here, paraphrase stayed best, contradiction was usually worst, and neutral remained strongly disruptive. The final group again showed larger drift than the initial group overall.

The combined interpretation is that OpenVLA is meaningfully language-sensitive rather than completely image-dominant. However, it is not perfectly stable under paraphrase, and its behavior becomes less aligned when the scene is already near a completed state. That

makes prompt wording and scene stage both important variables in practical VLA deployment.

Invariance and closest/farthest counts

Branch / group	Exact all-same	Near-all-same	Paraphrase closest	Contradiction farthest	Neutral farthest
manual40 initial	1	8	25	14	21
manual40 final	0	4	22	12	21
auto50 initial	0	1	27	15	22
auto50 final	0	1	26	16	15

These counts are useful for defense discussion. The manual40 branch contained a few near-invariant cases, especially in initial scenes, which suggests the model sometimes collapses to nearly the same action regardless of prompt family. The broader auto50 branch showed fewer such cases, which is one reason the second branch strengthens the paper rather than duplicating it.

Bootstrap 95% confidence intervals for L2 means

Experiment	Group	Metric	Mean	95% CI
manual40	initial	l2_normal_paraphrased	0.028126	[0.016002, 0.042951]
manual40	initial	l2_normal_contradictory	0.042365	[0.027787, 0.059539]
manual40	initial	l2_normal_neutral	0.045330	[0.032649, 0.059542]
manual40	final	l2_normal_paraphrased	0.043141	[0.030908, 0.056137]
manual40	final	l2_normal_contradictory	0.059920	[0.042288, 0.079732]
manual40	final	l2_normal_neutral	0.070772	[0.052691, 0.090367]
auto50	initial	l2_normal_paraphrased	0.056164	[0.041880, 0.071180]

auto50	initial	l2_normal_contradictory	0.069716	[0.053953, 0.087871]
auto50	initial	l2_normal_neutral	0.073405	[0.060928, 0.088257]
auto50	final	l2_normal_paraphrased	0.080668	[0.061054, 0.103965]
auto50	final	l2_normal_contradictory	0.093857	[0.076429, 0.112974]
auto50	final	l2_normal_neutral	0.082103	[0.068564, 0.097324]

8. Failure Analysis and What Went Wrong

- Cloud notebook environments were unstable for this project because model imports, bitsandbytes support, and remote-code dependencies did not line up cleanly.
- The first local model load failed due to missing psutil; that was easy to fix but proved that package completeness matters.
- The main OpenVLA failure was the attention-mask mismatch after vision fusion. This caused shape errors during generation until the mask was patched.
- The project mixed image folders at one point because the manual40 images and auto50 sampled images used the same traj### naming pattern. The solution was conceptual separation: manual40 was frozen as completed result CSVs, while the active image folder was treated as the auto50 branch.
- Some scripts were partially garbled by shell paste errors. These were fixed by re-creating clean script files rather than editing damaged ones line by line.
- The auto50 smoke-test results were briefly summarized from the wrong file. The fix was to use results/results_all_auto50.csv rather than the old smoke-test results_all.csv.

Why this section matters: a thesis submission should not hide the engineering failures. The debugging process shows methodological maturity, helps reviewers trust the artifact, and gives future readers a map of what to watch for when reproducing the work.

9. Research Gap Revisited

The final project closes a modest but real gap. Existing VLA work demonstrates strong capabilities at scale, but practical prompt sensitivity in control outputs is not yet a settled question. This project contributes evidence that action drift under language perturbation is measurable, that scene stage changes the effect size, and that the pattern appears in both a curated small study and a broader validation set. The work does not claim to solve VLA robustness, but it provides a concrete, reproducible lens for studying it.

10. Limitations

- Only OpenVLA-7B was tested. A stronger paper would compare at least one additional model such as a smaller VLA baseline or a newer efficient open model.
- The manual40 branch used task-specific prompt writing, while auto50 used family-derived prompts. This is a strength for breadth, but it means the two branches are not a pure size-controlled comparison.
- The project reported drift on the first six action dimensions and did not separately analyze the seventh dimension in depth.
- The project used static images from trajectories rather than full video-conditioned rollout evaluation. That makes this a prompt-sensitivity study on action prediction, not a full closed-loop deployment study.
- No human subject study was run to measure how people phrase instructions in the wild. That would be a strong follow-up.

11. Why the Project Is Still Strong

Despite these limitations, the project is strong enough for a serious report because the experiments are complete, the environment is reproducible, the results are internally consistent, and the work includes both methodology and failure analysis. It does not rely on a single lucky chart. It includes a completed manual study, a completed automatic validation study, confidence intervals, family breakdowns, logs, and a clear engineering narrative.

12. Future Work

- Add a second model baseline, ideally another open VLA or an efficient smaller policy, to test whether the drift pattern is OpenVLA-specific or more general.
- Run a controlled paraphrase set with multiple paraphrase strengths rather than a single paraphrase per prompt.
- Separate the seventh action dimension and analyze whether it behaves like a mode switch or stop signal in some final-state cases.
- Add family-level ablations and task-category visuals in the final paper appendix.
- Produce a short video showing one or two representative trajectories and the corresponding drift tables for supplementary material, since CoRL encourages supplementary zip and video material. [5]
- Package the code and results on GitHub with a clear README, then keep the 30 GB dataset out of the public repo and instead link to the official source. [2]

14. References

[1] Kim et al., "OpenVLA: An Open-Source Vision-Language-Action Model," Proceedings of CoRL, PMLR 270, 2025. (https://arxiv.org/abs/2406.09246)

[2] Walke et al., "BridgeData V2: A Dataset for Robot Learning at Scale," Proceedings of CoRL, PMLR 229, 2023. (https://arxiv.org/abs/2308.12952)

[3] Cadene et al., "LeRobot: An Open-Source Library for End-to-End Robot Learning," ICLR 2026 conference paper.(https://arxiv.org/abs/2602.22818)

[4] Sapkota et al., "Vision-Language-Action (VLA) Models: Concepts, Progress, Applications and Challenges," arXiv:2505.04769, 2026.(https://arxiv.org/abs/2505.04769)

[5] Conference on Robot Learning, "Instruction for Authors," official conference website, accessed April 2026. (https://www.corl.org/contributions/instruction-for-authors)

[6] IEEE Robotics and Automation Letters, "Information for Authors," official IEEE RAS website, accessed April 2026.(https://www.ieee-ras.org/publications/t-rl/information-for-authors/)